WATER RESOURCES, WELLS & PUMPS · NOTE WR01

# Irrigation and Water Resources — Hydrologic Cycle, Global and Indian Water Scenario, Groundwater Status

*Course topics: IDE5.5 L2, SWCE3.5 L2, IDE6.2 L3, IDE-506 T1*

## What irrigation is

Irrigation is the application of water to soil to help produce crops. It supplements the water a crop gets from rainfall, stored soil moisture and capillary rise from groundwater. In many parts of the world, the amount and timing of rainfall cannot meet crop needs, so successful cropping often depends on irrigation.

- Irrigation is as old as civilisation and goes back to Vedic times. An ancient Smriti text (cited in the notes as Narmada Smriti XI, 9 — probably the Nārada Smṛti) warns: "No grain is ever produced without water, but too much water tends to spoil the grain; an inundation is as injurious to crop growth as a dearth of water."
- Megasthenes, Greek ambassador at the court of Chandragupta (300 B.C.), recorded that the country prospered with two crops a year under irrigation, and that district officers measured the land and inspected sluices so that branch canals shared water fairly.

## Scope and objective

- **Scope:** from the watershed that yields the water, through its storage, conveyance, management and distribution, to the drainage problems and waterlogging caused by faulty irrigation — all are the irrigation engineer's concern.
- **Why it matters:** water is limited, and its efficient use is basic to feeding a growing world population.
- **Plan water development around:** availability, quality, location and distribution, variation in occurrence, climate, soil, competing demands and socio-economic conditions.
- **Objective of efficient irrigation:** raise agricultural production per unit volume of water, per unit cropped area and per unit time.

## Hydrology and the hydrologic cycle

Hydrology is the earth science of the occurrence and movement of water on and beneath the land. All precipitation comes from the earth's unending moisture cycle — the **hydrologic cycle** — in which water circulates between land, oceans and atmosphere.

1. Evaporation from oceans, lakes, rivers and land
2. Condensation in the atmosphere
3. Precipitation
4. Interception by vegetation
5. Infiltration into the soil
6. Surface runoff
7. Evaporation from soil and water bodies
8. Transpiration from plants

9. Percolation to groundwater
10. Groundwater flow to streams and oceans

Soil acts as a reservoir: water held there in transitory storage flows underground to streams, percolates to groundwater or returns to the air by evaporation. Eventually all of it re-enters the cycle.

## Global water scenario

| Item | Figure |
|---|---|
| Oceans / land share of the earth's surface | 70.8% / 29.2% |
| World water in oceans (saline) | About 97.5%; only 2.5% is fresh |
| Locked in ice caps and glaciers | About 2% of all water — most of the fresh water — generally unavailable |
| Groundwater deeper than 800 m | About half of all groundwater; not ordinarily available for economic development |
| Water available for development | Streams, rivers and fresh-water lakes, plus about half of the shallow groundwater (less than 800 m deep) |

- Global water use has grown six times in the past century, and demand is growing twice as fast as population.
- At least four billion people face water scarcity for at least one month every year.
- Contaminated water is often quoted as causing about 80% of the disease burden in developing nations.
- Diarrhoea is a leading killer of young children — the notes call it the second leading cause of death for children under five. **Update:** the WHO fact sheet of 7 March 2024 ranks it the **third** leading cause of death in children 1–59 months old (about 443,832 deaths a year).

| Category (per capita per year) | Threshold |
|---|---|
| Water-stressed | < 1700 $m^3$ |
| Water-scarce | < 1000 $m^3$ |
| Absolute water scarcity (AWS) | < 500 $m^3$ |

## India's water scenario

- India has 17.76% of the world's population but only 4% of its fresh water.
- Water is unevenly spread: the east and north have more, the south and west less.
- Climate change is increasing droughts, floods and rainfall variability.

| Year | Per-capita water availability ($m^3$/year) |
|---|---|
| 1951 | 5177 |
| 2011 | 1545 |
| 2021 | 1486 |
| 2031 (projected) | 1367 |

India is on the verge of becoming water-scarce — it is already below the 1700 $m^3$ water-stress line.

*The 2021 and 2031 values (1486 and 1367 m³) were confirmed in a Rajya Sabha reply in December 2022. The Central Water Commission's 2019 reassessment raised India's average annual water resources from 1869 BCM to **1999.20 BCM**, with utilisable water of about 1137 BCM (690 BCM surface + 447 BCM groundwater).*

## Future water scenario

| Estimate | Value |
|---|---|
| Utilisable water — Ministry estimate | 1123 BCM |
| Utilisable water — T. N. Narasimhan (2008) | 712 BCM |
| Annual requirement 2025 / 2050 | 843 BCM / 1180 BCM |

- By 2025, one-third of India's population may face absolute water scarcity.
- By 2030, demand may reach 1.5 trillion m³.
- World Bank (2012): 60% of aquifers may become critical.
- Mihir Shah Committee (2016): 50% of water demand may go unmet by 2030.

## India's groundwater

- Total replenishable groundwater: 432 km³, about 46% of it in the Ganga–Brahmaputra basin.
- Groundwater irrigates more than half of the irrigated land and supports about 20 million tubewells.

| CGWB assessment 2023 | Value |
|---|---|
| Annual recharge | 449.08 BCM |
| Annual extractable resource | 407.21 BCM |
| Annual extraction | 241.34 BCM |
| Stage of extraction | 59.26% |
| Units: over-exploited / critical / semi-critical | 11.23% / 3.04% / 10.65% |
| Units: safe / saline | 73.14% / 1.94% |

*Update — CGWB's 2025 national assessment (as reported): recharge 448.52 BCM, extractable resource 407.75 BCM, extraction 247.22 BCM, stage of extraction 60.63%; 730 of 6762 assessment units (10.8%) over-exploited. Check the CGWB/PIB release before quoting.*

**Rajasthan focus (Dynamic Ground Water Resources of Rajasthan 2024, CGWB & State GWD):** 302 assessment units (295 blocks + 7 urban). Recharge 12.58 BCM (9.30 BCM from rain, 3.28 BCM from other sources); extractable 11.37 BCM; extraction 17.05 BCM; **stage of extraction 149.86%**. 214 units (70.86%) over-exploited, 27 (8.94%) critical, 21 (6.95%) semi-critical, 37 (12.25%) safe, 3 (0.99%) saline. Average rainfall in 2023 was 695 mm — under 400 mm in western Rajasthan, with over 75% falling from June to September.

## Water budgeting

Water budgeting is like balancing a bank account: rainfall is the saving, losses and uses are the withdrawals. It estimates the water harvested and used over a set period.

- **Objectives:** assess the water available and the water required; find the deficit or surplus; prepare action plans.
- **Data needed:** total watershed area and its distribution; cropping pattern and land use; soil and water conservation structures; rainfall (average and actual); population and livestock.

- **Practice:** the ideal unit is the micro-watershed, but it is practically worked out at village level, twice a year — pre-monsoon (kharif) and post-monsoon (rabi and summer). It is not a one-time exercise.

> **Narasimhan and Gaur (2009):** average rainfall 1170 mm gives a total rainfall yield of 3838 km³; evapotranspiration takes 40%; utilisable water is 1123 km³. On these figures, demand had not yet exceeded supply.

**Sources:** Central Water Commission, Reassessment of Water Availability in India using Space Inputs (2019); Rajya Sabha reply (Dec 2022) on per-capita availability; CGWB Dynamic Ground Water Resources of India (2023 and 2025 assessments); CGWB & Ground Water Department, Govt of Rajasthan, Dynamic Ground Water Resources of Rajasthan 2024; WHO fact sheet 'Diarrhoeal disease' (7 March 2024). ICAR eCourse 'Irrigation Engineering' — Dr K. N. Tiwari and Dr N. S. Raghuwanshi (IIT Kharagpur), reviewed by Dr S. D. Gorantiwar (MPKV Rahuri). Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

## Practice MCQs

**Q1. A country is water-stressed when per-capita availability falls below**

(a) 500 m³/yr (b) 1000 m³/yr
(c) 1700 m³/yr (d) 2500 m³/yr

**Q2. India's per-capita water availability in 2021 was assessed as about**

(a) 5177 m³ (b) 1545 m³
(c) 1486 m³ (d) 1000 m³

**Q3. Stage of groundwater extraction for India in the 2023 assessment was**

(a) 39.5% (b) 59.26%
(c) 75.1% (d) 149.86%

**Q4. Rajasthan's stage of groundwater extraction in the 2024 assessment was about**

(a) 59% (b) 96%
(c) 150% (d) 200%

**Q5. About what share of the world's water is fresh?**

(a) 0.5% (b) 2.5%
(c) 10% (d) 29.2%

**Q6. The ideal unit for water budgeting is the**

(a) river basin (b) micro-watershed
(c) district (d) state

**Answers**

Q1 — (c) 1700 m³/yr
Q2 — (c) 1486 m³
Q3 — (b) 59.26%
Q4 — (c) 150% — *149.86%; 214 of 302 units over-exploited.*
Q5 — (b) 2.5%
Q6 — (b) micro-watershed — *In practice it is done at village level.*